



Technical Proposal

What we will build, and with what

AWNAK — Home & Family Services Platform

The full engineering specification: system architecture, the technology stack and why each part was chosen, the data model, the dispatch engine, the order lifecycle, the money layer and commission rules, external integrations, and the security, performance and scalability standards — with the phase-one scope stated precisely.

Commercial in confidence

DOCUMENT

RAX-TEC-AWNAK-001-EN

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Prepared for the owner of the Awnak project, against the home-and-family services requirements specification. Read together with RAX-FIN-AWNAK-001-EN — the Commercial Proposal, Ways of Working and Delivery.

Contents

1	Executive summary	13	Mobile app — the customer journey
2	Understanding the project and scope boundaries	14	Provider mode and provider dashboard
3	System architecture	15	Admin panel and permissions
4	Technology stack and the reasoning behind it	16	Reporting and export engine
5	Data model and database	17	Marketing website
6	Matching and dispatch engine	18	External integrations
7	Order lifecycle and pricing models	19	Security and data protection
8	Payments, wallet and commission	20	Performance and scalability
9	In-app messaging and notifications	21	Reliability, backup and monitoring
10	Search, filtering and Arabic text handling	22	Testing and quality assurance
11	Maps and location	23	Infrastructure, deployment and operations
12	Visual identity and design system	24	Readiness for future expansion
		25	Technical deliverables at handover

How to read this document

This document answers one question: **what will be built, with what technology, and within what boundaries**. Prices, the payment plan, durations, ways of working, delivery and warranty are in the second document, RAX-FIN-AWWAK-001-EN. The two are complementary and neither should be read in isolation.

1 Executive summary

What is required is not a directory of tradespeople, nor a classified-ads platform, but a complete service operations and management platform — beginning with a customer's need and ending with a financial report. That understanding shaped every engineering decision in this document.

We read the requirements specification in full, across all thirty-two of its sections, and built an interactive prototype of the customer journey which was approved visually before this proposal was written. Our conclusion is that the success of **AWNAK** rests on four engineering pillars, not on a screen count:

Service categories that flex without code

The specification states explicitly that the operator must be able to add new categories and services **without any programming change**. The catalogue is therefore built on a dynamic field schema defined from the admin panel, not on fixed tables per trade.

A dispatch engine, not a search list

An order is not merely published — it is routed against ten criteria: specialisation, location, coverage radius, rating, response speed and availability. That is a scoring engine whose weights are tunable from the panel.

A ledger, not a "price" field

Held funds, multi-rule commission, additional costs, partial refunds and provider balances all require **double-entry bookkeeping**. Anything less makes the balances impossible to audit later.

Arabic-first, not translated

Native RTL layout, search that tolerates variant spellings of hamza and taa marbuta, grammatical gender that follows the account, and type sizing that respects older users — handled in the search layer and the interface together.

1.1 Phase one in numbers

1

unified app
customer + provider

2

platforms
Android and iOS

6

pricing models
fully supported

15

order states
in the state machine

11

main categories
with sub-services

15

approved screens
in the customer journey

9

external integrations
each priced separately

24

admin reports
all exportable

1.2 A foundational architectural decision

One account, two modes — not two accounts

As the specification requires, identity (**User**) is separated from the professional profile (**Provider Profile**) in the data model. A user remains a single user and a professional profile is attached after verification. This prevents duplicated records, makes switching between customer mode and provider mode instantaneous without signing out, and keeps order history and ratings tied to one identity.

2 Understanding the project and scope boundaries

2.1 The four project components

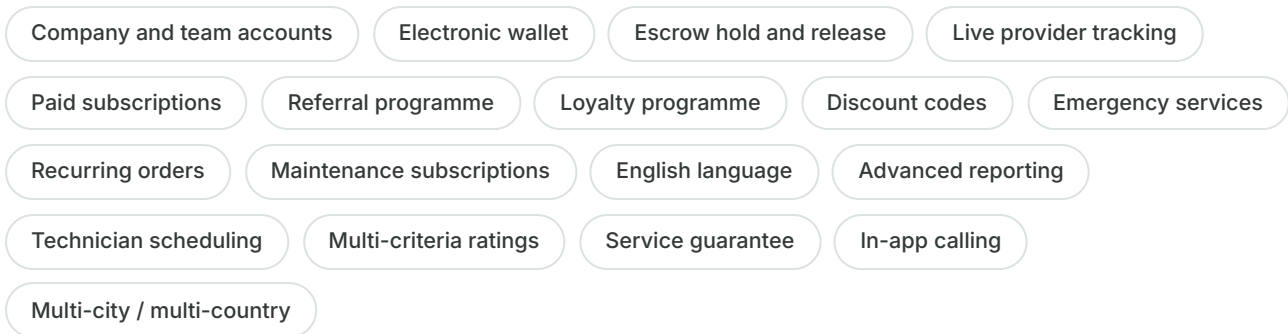
Component	Description	Phase
Mobile application	A single app on Android and iOS, allowing a switch between customer mode and provider mode without two separate accounts.	Phase 1
Admin panel	A web application to manage customers, providers, orders, quotes, payments, commissions, complaints, ratings, notifications, content, reports, coverage areas, service categories, and staff with their permissions.	Phase 1
Marketing website	A landing page covering the app introduction, service types, how to order, how to register as a provider, download links, terms, privacy policy, FAQ and a contact form.	Phase 1
Backend system	Backend server, database, APIs, media handling, geolocation, notifications, payments, permissions, logs and reporting.	Phase 1

2.2 Functionality included in phase one

- ✓ Customer registration by phone with OTP
 - ✓ Provider registration and document verification
 - ✓ Dynamic service category management
 - ✓ Location and multiple saved addresses
 - ✓ Creating a service order with photos and video
 - ✓ Dispatching the order to matching providers
 - ✓ Submitting and comparing quotes
 - ✓ Selecting a provider and closing quoting
 - ✓ In-app messaging
 - ✓ Tracking the fifteen order states
 - ✓ Cash payment on completion
 - ✓ One electronic payment gateway
- ✓ Manageable commission calculation
 - ✓ Two-way ratings
 - ✓ Complaints and dispute handling
 - ✓ Push notifications
 - ✓ Customer and provider profiles
 - ✓ Portfolio of previous work
 - ✓ Complete admin panel
 - ✓ Core reports and export
 - ✓ Arabic language with RTL support
 - ✓ Marketing website
 - ✓ Publishing to Google Play and the App Store
 - ✓ Admin training and source code handover

2.3 Functionality deferred to phase two

Outside the phase-one scope and priced item by item in the commercial document. Each has been accounted for architecturally so that enabling it later requires no rebuild:

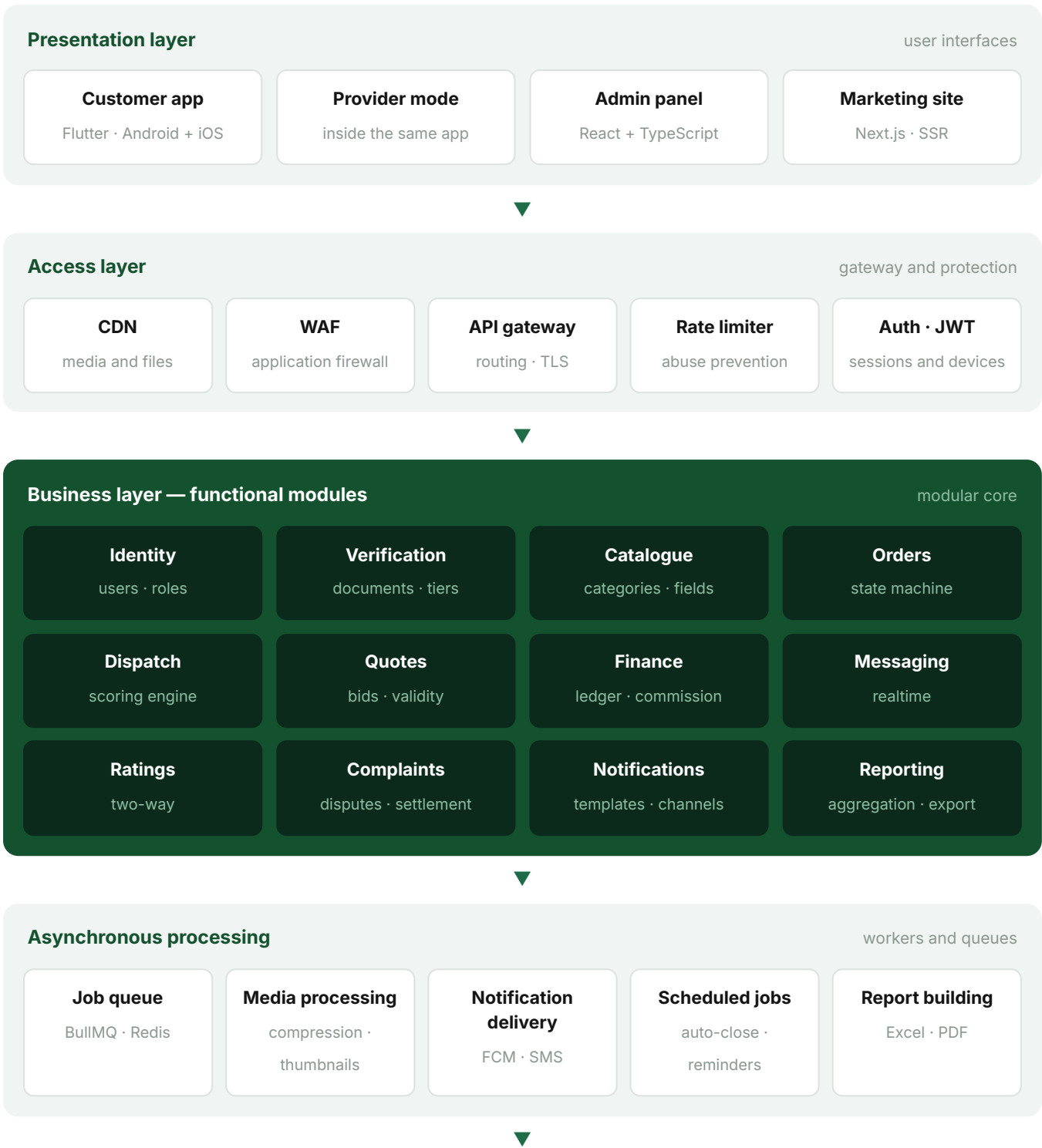


Scope boundaries

The items listed in section twenty-seven of the specification — purchasing hardware or servers; payment gateway, messaging, maps and app store fees; marketing photography; provider data entry; the call centre; advertising campaigns; visual identity; professional translation; and insurance — are **not included** unless stated explicitly in the commercial proposal. The full detail is in section 8 of the commercial document.

3 System architecture

A layered architecture with modular components inside a single separable core — not distributed microservices. The decision is deliberate and explained in 3.2.



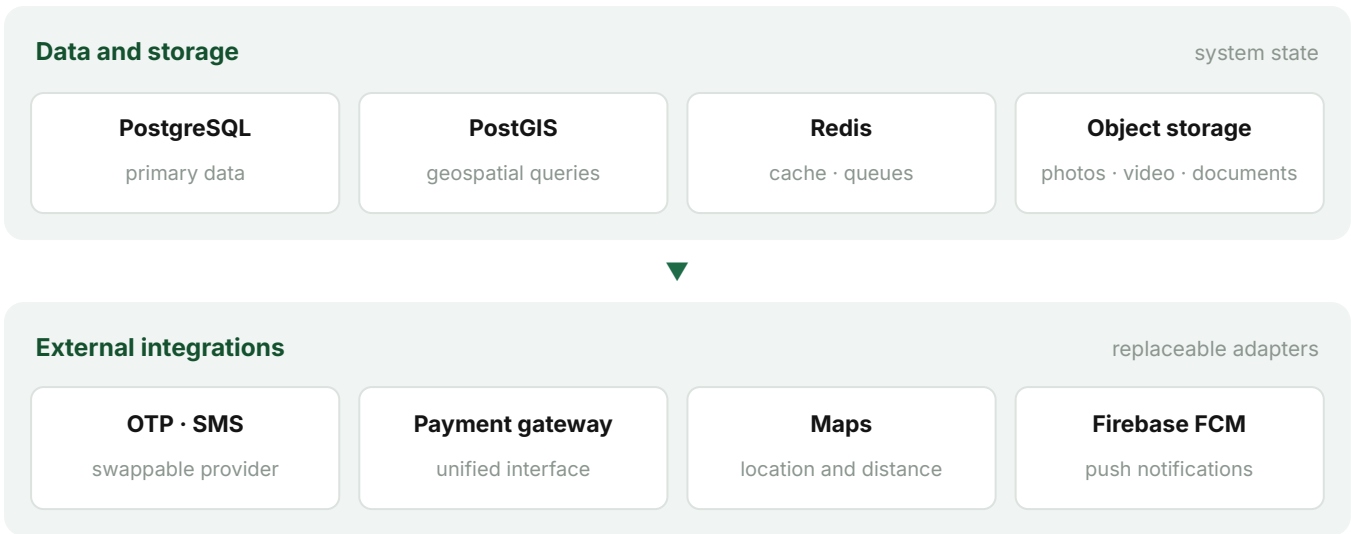


Figure 1 — The six system layers. The business layer is the modular core; above it are interfaces, below it are infrastructure and integrations that can be replaced without touching business logic.

3.1 The replaceable adapter principle

Every external integration sits behind a **single internal interface**. Changing the SMS provider or adding a second payment gateway requires writing a new adapter only, not editing business logic — which serves the "ability to add new payment gateways" requirement directly, and protects the project from being locked to any one vendor.

3.2 Why a modular core rather than microservices

Criterion	Decision and reasoning
Financial transaction integrity	Holding funds, deducting commission and releasing the provider's balance must occur in a single atomic transaction. Splitting this across microservices imposes complex and risky distributed transactions on a financial system.
Team size and delivery time	Microservices impose an infrastructure overhead that consumes phase-one budget without delivering value before a large user base exists.
Separability later	Modules are separated by clear boundaries and internal interfaces, so any module — messaging or notifications, for example — can be extracted as a standalone service when needed, without a rewrite.
Horizontal scaling	The core is stateless and runs as multiple instances behind a load balancer; asynchronous workers scale independently of the API.

4 Technology stack and the reasoning behind it

4.1 Mobile application

Element	Technology	Reasoning
Framework	Flutter 3.x · Dart	The specification asks for a single app running on Android and iOS. One codebase with native performance, mature built-in RTL support, and full control over fonts and their sizing — an explicit requirement for accommodating older users.
State management	Riverpod + Freezed	Order state passes through fifteen stages and changes in real time; this needs a strict state model that makes invalid states unrepresentable at the type level.
Local storage	Drift (SQLite) + Secure Storage	To meet "the rest of the app must not stop on a weak connection": orders and messages are stored locally and synchronised when the network returns. Sensitive tokens live in encrypted storage.
Networking	Dio + interceptors	Automatic retry, token refresh, and an upload queue for media.
Maps	Google Maps SDK	Location, manual address selection, distance calculation, coverage display.
Realtime	Socket.IO client	Messaging, order status updates and incoming quotes in real time, without battery- draining polling.
Media	image_picker · video_compress	Meets "compress photos and video before upload" — compression happens on device.

4.2 Backend

Element	Technology	Reasoning
Framework	NestJS · Node.js 22 (TypeScript)	A module and dependency-injection structure that translates domain boundaries directly into code, with excellent performance under I/O-heavy load. TypeScript gives type safety shared between the server and the panel.

Element	Technology	Reasoning
APIs	REST + OpenAPI 3.1	Automatic documentation and client generation — and technical documentation is an explicitly required deliverable.
Realtime	Socket.IO + Redis adapter	Works across multiple server instances.
Queues	BullMQ on Redis	Notifications, media processing and report building run outside the request cycle.
Authentication	JWT + refresh rotation	Bounded session length, session revocation and active device management — explicit security requirements.
Authorisation	RBAC + CASL	Fine-grained permissions per staff member, and restricted access to financial reports.
Validation	Zod / class-validator	Strict validation of every input at the API boundary.

4.3 Data and storage

Element	Technology	Reasoning
Database	PostgreSQL 16	Full ACID transactions — a precondition for the financial system — plus JSONB for dynamic fields and built-in full-text search.
Geospatial	PostGIS	For "dispatch by coverage radius" and "distance calculation": spatial indexes answer "who covers this point" in a fraction of a second.
Cache	Redis 7	Sessions, provider availability state, rate limiting, queues and event publishing.
Media storage	S3-compatible + CDN	Portfolio images, documents and video, with short-lived signed URLs for sensitive documents.
Schema migration	Prisma Migrate	Versioned, reviewed migrations, and a test environment matching production.

4.4 Admin panel and website

Element	Technology	Reasoning
Admin panel	React 18 + TypeScript + Vite	An internal panel needs no search-engine indexing; a single-page app is faster for large tables and filtering.
UI library	MUI (RTL) + TanStack Table	Native RTL support, and large tables with server-side pagination and advanced filters.
Charts	Recharts	The KPI dashboard, growth and geographic distribution.
Marketing website	Next.js (SSR/SSG)	For SEO and load speed: server-rendered pages with correct Arabic meta tags.

4.5 Infrastructure and operations

Element	Technology	Purpose
Containers	Docker + Compose	Identical environments across development, test and production.
Continuous integration	GitHub Actions	Automated build, test and deploy with quality gates.
Reverse proxy	Nginx + TLS	TLS termination, compression, security headers.
Error tracking	Sentry	Meets "fault monitoring and error logging".
Logs and metrics	Prometheus + Grafana + Loki	Performance monitoring and alerting.
Backups	Managed snapshots + log archiving	Daily backups with periodic restore testing.
Analytics	Firebase Analytics	In-app user behaviour.

5 Data model and database

A normalised model that separates **identity** from the **professional profile** from the **financial entry**. That separation is what allows one account with two modes, sound financial auditing, and adding new services without a code change.

5.1 Principal table groups

Group	Main tables	Design notes
Identity and access	users, user_devices, sessions, otp_challenges, roles, permissions, admin_users, audit_log	A single user record. Roles are separated to allow multiple roles. The audit log records every sensitive administrative action.
Professional profile	provider_profiles, provider_documents, provider_services, provider_coverage, provider_schedule, portfolio_items	An optional professional profile attached to the user. Coverage is stored as PostGIS geometry and the schedule as queryable weekly intervals.
Catalogue	categories, services, service_fields, pricing_models, service_requirements	The flexibility pivot: <code>service_fields</code> defines each service's fields as data rather than code, so a new trade is added from the panel.
Locations	governorates, cities, districts, user_addresses, coverage_zones	A full hierarchy — governorate, city, district, neighbourhood — with travel fees per area.
Orders	orders, order_items, order_media, order_status_history, order_extras, dispatch_log	Every state transition is recorded with its timestamp and actor — the basis for the stalled-orders and response-time reports.
Quotes	quotes, quote_lines, quote_revisions	A quote carries labour, materials, duration, expected arrival time and validity period.
Finance	ledger_entries, wallets, payments, refunds, commission_rules, payouts, invoices	An immutable double-entry ledger. Any correction is posted as a reversing entry rather than by editing the record.
Communication	conversations, messages, message_flags, notifications, notification_templates	Conversations are tied to the order and retained as evidence in disputes.
Quality	reviews, review_criteria, complaints, complaint_messages, disputes	Two-way ratings restricted to a genuinely completed order, one per party per order.
Content	banners, pages, faqs, settings	Content editable from the panel without publishing a new release.

5.2 The architectural key — dynamic service fields

The hardest requirement in the specification is that the operator can add new services **without a programming change**. The answer is to define each service's order form as data:

Field definition

For each service the operator defines its fields: type (text, number, choice, image, date), whether it is mandatory, its permitted values, and display order.

Automatic rendering

The app builds the order screen from that definition at runtime, so a new service appears with the correct form without an app store update.

Flexible but indexed

Answers are stored in an indexed JSONB column, so they remain searchable, filterable and reportable despite the flexibility.

What this means in practice

Adding a new trade — "curtains and upholstery technician", say — with its own fields, pricing model, required documents, commission and coverage areas is done **from the admin panel in minutes**, with no new app release and no waiting for store review.

5.3 Data policies

- **Soft deletion** for every record tied to a financial transaction or a dispute, to preserve auditability.
- **Column-level encryption** for identity numbers and financial data.
- **Document isolation** in dedicated storage behind short-lived signed URLs, reachable only by an authorised staff member, with every access recorded in the audit log.
- **Deletion on request** under the privacy policy, anonymising financial entries rather than deleting them, so the books stay sound.

6 Matching and dispatch engine

The specification sets out ten criteria for dispatching an order. We implement them as a scoring engine whose weights are **tunable from the admin panel**, not fixed in code.

6.1 Engine stages

- 1

Hard filtering
Accounts failing a mandatory condition are excluded outright: mismatched specialisation, location outside the coverage radius, unverified or suspended account, outside declared working hours, or availability toggled off. This runs as a single PostGIS query against a spatial index.
- 2

Weighted scoring
Each remaining candidate is scored against the soft criteria in the table below. Weights are set from the panel and tested before being adopted.
- 3

Staged fan-out
The order goes first to the highest-scoring tier; if not enough quotes arrive within a set window, the circle widens automatically. This rewards the best performers without leaving a customer's order without quotes.
- 4

Throttling
A cap on orders sent to a provider per period, and a cap on quotes per order — protecting quote quality and the customer's experience.
- 5

Verification and subscription gating
Verification tier and subscription plan affect priority and the monthly quota of quotes, supporting the revenue model later without a code change.

6.2 Scoring criteria and default weights

Criterion	Initial weight	How it is calculated
Geographic proximity	20%	Linear decay with distance within the declared coverage radius.
Overall rating	20%	Weighted average giving greater weight to recent ratings.
Response speed	15%	Median time from order receipt to quote submission over the last thirty days.

Criterion	Initial weight	How it is calculated
Completed orders	12%	Logarithmic scaling to prevent incumbents monopolising dispatch.
Verification tier	12%	Registered < verified < verified company.
Order acceptance rate	8%	Accepted ÷ dispatched, above a minimum sample size.
Punctuality	7%	Share of arrivals within the quoted arrival window.
Availability at the requested time	6%	Declared schedule against the requested service time.
Years of experience	6%	A supporting factor with a defined ceiling.
Cancellation rate	-6%	A penalty for provider-side cancellations.

These weights are initial and will be confirmed with the project owner during the analysis phase. They remain adjustable from the admin panel at any time without a new release.

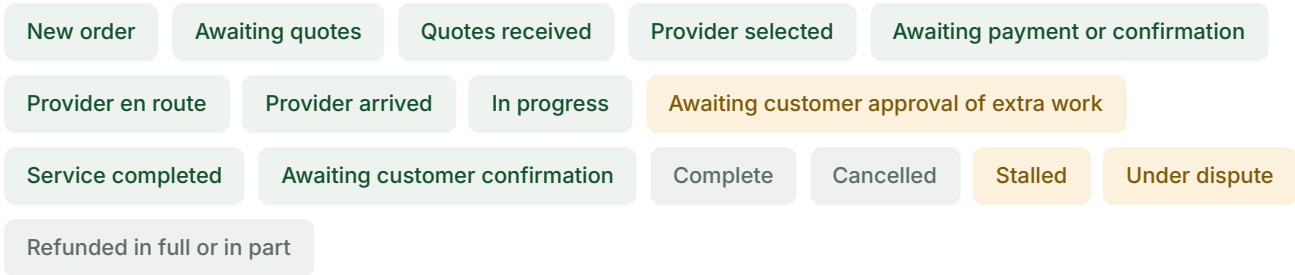
An explicit requirement, addressed

The specification warns that subjecting every order to manual approval delays urgent work. An order is therefore published **automatically** once it passes automated checks — data completeness, location validity, prohibited content screening, banned keywords, duplicate detection and account eligibility — and only what the rules classify as **suspicious** is routed to manual review.

7 Order lifecycle and pricing models

7.1 The order state machine

The fifteen states are implemented as an explicit state machine: every transition has a condition and an authorised actor, and is recorded in `order_status_history` . Disallowed transitions are rejected at the system level, not merely hidden in the interface.



7.2 The seven stages of an order

#	Stage	What the system does
1	Order creation	Category and service selection, a short title, a description of the problem, photo and video attachments, service address and map location, timing (now / today / a later slot), urgency level, an optional indicative budget, and preferred payment method.
2	Order screening	Automated checks on data completeness, location validity, service type, prohibited content, banned keywords, duplicates and account eligibility. Publication is automatic; only suspicious orders go to manual review.
3	Dispatch	The matching engine (section 6) routes the order across staged tiers.
4	Quotes	Proposed price, labour charges, materials cost, duration, expected arrival time, quote validity period, conditions, notes and attachments.
5	Selection	The customer compares quotes — name, photo, verification badge, rating, jobs completed, experience, price, arrival time, duration. On selection, quoting closes, the chosen provider is notified, and the others are told the competition has ended.
6	Execution	State progression, additional work or materials only with prior customer approval, and live status updates on both sides.
7	Completion	The provider marks the service complete and uploads after photos; the customer is notified and confirms; commission is deducted and the balance released; both parties are asked to rate. If the customer does not respond within a window set by the operator, the order closes automatically after a status review.

7.3 The six pricing models

The specification notes that services cannot all be priced the same way. Each model is implemented as an independent strategy, bound to a service from the admin panel:

Model	Used for	Behaviour in the system
Fixed price	Tyre inflation, tyre change, air-conditioner cleaning, light fitting, car wash, inspection visit.	The base price is shown before ordering; the quoting stage is skipped and the order is assigned directly.
Quotation	House painting, furniture moving, leak repair, appliance repair, carpentry and metalwork.	The full cycle: details and photos → competing quotes → selection.
Inspection	Work that cannot be estimated remotely.	An inspection visit fee is paid first — and deducted from the job value — then a final price is submitted after inspection and approved inside the same order.
Hourly	Hourly labour and open-ended work.	Total = hours × number of workers × hourly rate, with start and end times recorded.
Emergency	Urgent faults.	Dispatched to providers available right now within a tight radius, with an urgency surcharge set by the system or the provider under platform policy.
Direct request	Ordering from a specific provider's page, or repeating a past order.	Bypasses dispatch and goes to the named provider, with the timing, address or details editable.

8 Payments, wallet and commission

The governing principle of the financial layer

Every movement of money is an **entry in an immutable ledger**. Balances are computed from the sum of entries rather than stored in a single field, and no entry is edited after posting — corrections are made by reversing entry. This is a precondition for auditable financial reporting and for resolving disputes.

8.1 Supported payment methods

Method	Phase	Notes
Cash on completion	Phase 1	Commission is posted as a debt against the provider, settled from later balances.
Electronic payment via one gateway	Phase 1	The gateway is chosen with the project owner; the integration is priced separately.
Electronic wallet and balance	Phase 2	The ledger foundation ships in phase one; the user-facing surface is added later.
Escrow hold	Phase 2	Detailed in 8.2; requires a gateway supporting hold and release.
Partial or full prepayment	Phase 2	Built on the wallet and the escrow hold.

8.2 The escrow cycle

- 1

Payment
The customer pays the service value electronically after accepting a quote.
- 2

Hold
The amount is posted to the platform's trust account and not transferred to the provider; it appears to them as a "held balance".
- 3

Confirmation
The customer confirms completion, or the order closes automatically after the defined window.
- 4

Settlement
Platform commission is deducted by entry and the net added to the provider's available balance by matching entry.

5

Withdrawal

The provider requests a withdrawal; the operator reviews it and the transfer is executed and posted.

8.3 The commission engine

The specification states explicitly: **the commission rate must not be hard-coded; it must be managed from the admin panel.** We implement this as a prioritised rule table, applying the most specific matching rule:

Rule dimension	Value types	Example
Service type	Percentage	12% on cleaning services
Governorate	Fixed amount	
	Percentage with a floor	10% in a newly launched governorate
Provider	Percentage with a cap	
	Percentage with both	8% for a contracted premium account
Company		A negotiated rate for companies
Subscription plan		Lower commission on paid plans
Promotion		0% on the first three orders
Payment method		A differential offsetting gateway fees

8.4 Additional costs during execution

No additional work begins without the customer's explicit approval. The order moves to **"Awaiting customer approval of extra work"** and no increase is counted until approval. The provider adds a line with its reason and value, the customer is notified, and on approval the value is added to the order total and commission is recalculated on the new total.

8.5 The electronic invoice

Order number	Customer details	Provider details	Service description	Base price	Materials
Extras	Discounts	Commission	Taxes and fees	Payment method	Payment status

8.6 Cancellation and refunds

The cancellation policy is a **manageable matrix**: for each cancellation case and each service type, the financial outcome is set from the panel.

Cancellation case	Configurable outcomes
Before any quote is accepted	Cancellation fee · visit compensation · partial deduction · full refund · partial refund — the operator sets the rule for each intersection of case and service type.
After a quote is accepted	
Before the provider sets off	
After the provider sets off	
After arrival	
During execution	
Customer no-show	
Provider no-show	

9 In-app messaging and notifications

9.1 In-app messaging

Capabilities

- Instant text messages over WebSocket
- Sending images and locations
- Voice notes — optional, subject to approval
- Sending an updated quote inside the conversation
- Message delivery and read notifications
- Conversations retained against the order

Required controls

- **Phone number blocking** before a quote is accepted — a pattern filter that catches numbers written as digits, words or with separators
- Reporting abusive messages, with escalation to the operator
- Revealing the number after acceptance, or keeping contact inside the app — **controlled by a setting** the operator chooses
- Conversations are admissible evidence in disputes

9.2 The notification system

A **template and event** driven engine: every system event fires a template, and the template determines the channels and content. Templates are edited from the panel without a release.

Channel	Phase	Use
In-app notifications	Phase 1	A notification centre with read state.
Push notifications	Phase 1	The primary channel for real-time events, via FCM.
SMS	Phase 1	OTP codes and critical events only.
Email	Phase 1	Invoices, reports and administrative alerts.
WhatsApp Business	Optional	Priced separately, subject to service availability.

Notification-triggering events

- | | | |
|------------------------------|--------------------------|---------------------|
| ● Login code | ● Quote rejected | ● Service completed |
| ● Provider approved | ● Order cancelled | ● Payment confirmed |
| ● Verification rejected | ● Provider set off | ● Refund issued |
| ● New service order received | ● Provider arrived | ● Message received |
| ● Quote received | ● Work started | ● Complaint opened |
| ● Quote accepted | ● Approval of extra cost | ● Complaint updated |

- Rating requested
- Offers and discounts
- Subscription expiry
- Account suspended or reactivated

User control

The system separates **operational notifications** — which cannot be switched off because they are part of the service — from **marketing notifications**, which the user controls from their settings, as the specification explicitly requires.

10 Search, filtering and Arabic text handling

The specification requires search in Arabic that handles minor spelling variation. This is not a peripheral feature — it is the difference between a user finding a service and leaving the app.

10.1 The Arabic normalisation layer

Treatment	Example	Effect
Hamza normalisation	ا → آ / إ / أ	"إصلاح" and "اصلاح" return the same results.
Taa marbuta normalisation	ة → ة	"سباكة" and "سباكه".
Alif maqsura normalisation	ي → ي	"مبنى" and "مبنى".
Stripping diacritics and tatweel	vowel marks and kashida	Ignored in matching.
Digit normalisation	٠١٢٣ → 0123	Search works with Arabic and Latin digits alike.
Typo tolerance	Trigram similarity	"كهريجي" matches "كهريائي" by trigram similarity.
Synonyms	"مكيّف" ≡ "إسليت"	A synonym dictionary editable from the panel.

Implemented with `unaccent` and `pg_trgm` in PostgreSQL against an indexed normalised column — no external search engine is needed in phase one.

10.2 Filters available to the customer

Service name

Category

Governorate

City

District

Distance

Rating

Price

Verification tier

Arrival speed

Available now

Companies only

Individuals only

Emergency service

Fixed price

Favourites

11 Maps and location

Function	Phase	Implementation
Automatic customer location, with consent	Phase 1	Explicit permissions with a stated reason.
Manual location selection on the map	Phase 1	A draggable pin with reverse geocoding.
Multiple saved addresses	Phase 1	Home, work, parents' home, a family member's home, car location.
Distance calculation between parties	Phase 1	PostGIS for filtering, the maps API for real arrival time.
Dispatch by coverage radius	Phase 1	Coverage geometry per provider on a spatial index.
Approximate arrival time	Phase 1	From the routing API with caching to reduce consumption.
Operating areas set from the panel	Phase 1	Enable or disable any area, or any service within an area.
Live provider tracking once en route	Phase 2	Periodic location broadcast with privacy controls and automatic stop on arrival.

A privacy control written into the specification

The provider's precise location is not shown before the order is accepted, and full personal contact details are not shown before they accept. Until then only an approximate area and a distance band are displayed — never coordinates.

12 Visual identity and design system

The design system was approved as part of the interactive prototype of the customer journey, and is the binding visual reference for development. Its core tokens follow.

12.1 Colour palette

Primary #147D73	Primary dark #0F5F58	Deep ink #0B1F2E	Light tint #E6F2F1	Alert #F4B740	Success #12734B
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12.2 Typography and sizing

Element	Value	Note
Interface typeface	Tajawal	A clear geometric Arabic face supporting weights from Light to Black.
Companion Latin face	Inter	For Latin numerals and technical terms.
Base text size	16 pt minimum	To accommodate older users.
Touch targets	48 pt minimum	Reduces mis-taps.
Contrast	WCAG AA	Verified across every text and background combination.
Reference screen size	390 × 844	With a responsive layout from 4.7" to tablets.
Numerals	Arabic-Indic · ١٢٣	With search and input supported in both forms.

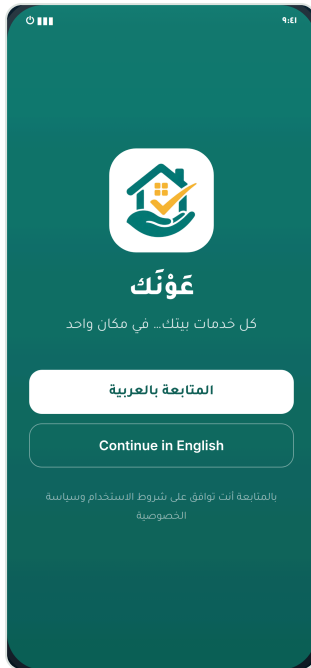
12.3 Grammatical gender

A linguistic detail settled in the prototype

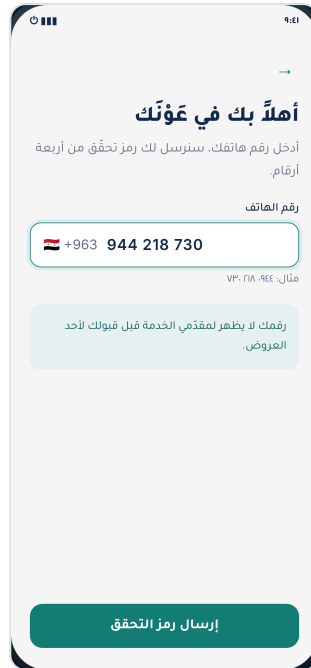
Interface copy is written in the **masculine** form by default and switches automatically to the **feminine** form when the account's gender is female. This requires strings to be stored in two forms in the translation files rather than one — which has been accounted for in the internationalisation structure since design.

13 Mobile app — the customer journey

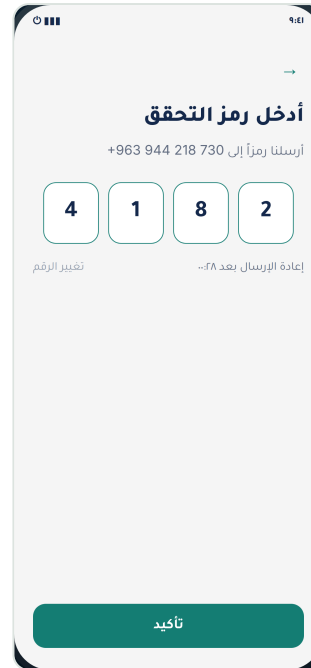
Fifteen screens covering the complete customer journey, from sign-in to rating. The screens below are from the approved interactive prototype and are the binding visual reference for development.



01 - Sign-in and language
Language chosen before any step



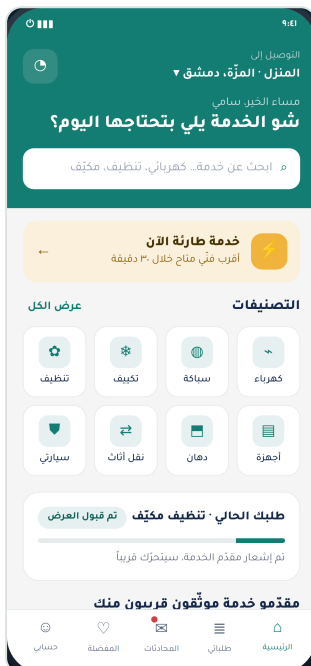
02 - Phone number
Entry and code dispatch



03 - Verification code
Resend timer



04 - Details and permissions
Location and notifications



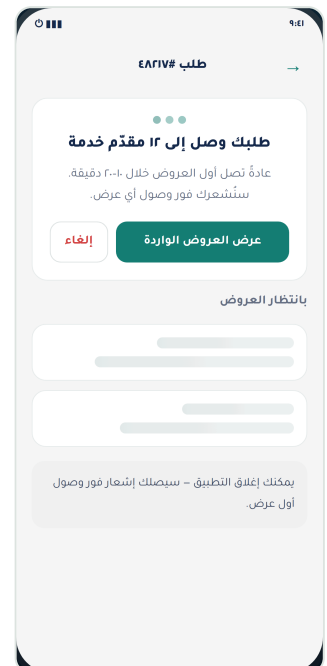
05 - Home
Categories and emergency service



06 - Categories and service
Browsing and service detail



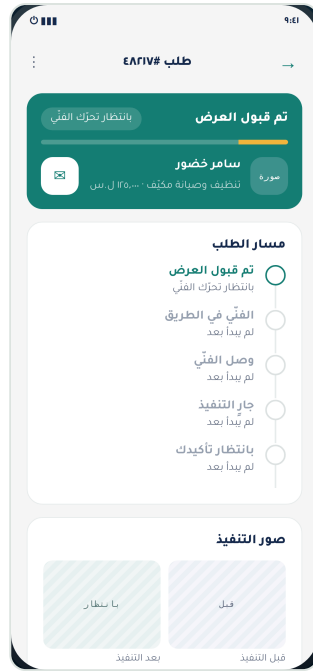
07 - Order wizard
Description, photos and timing



08 - Awaiting quotes
Dispatch status



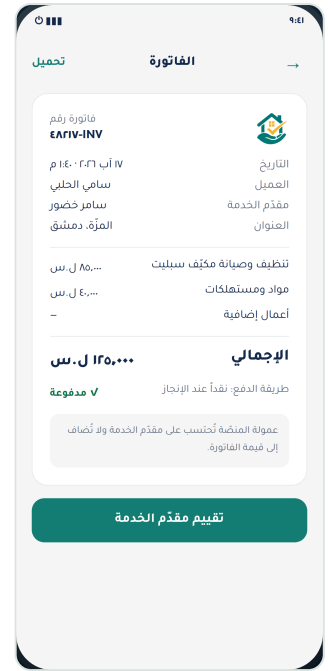
09 · Quotes and comparison
Comparing providers



10 · Execution tracking
Order state progression



11 · Messaging
Communication within the order



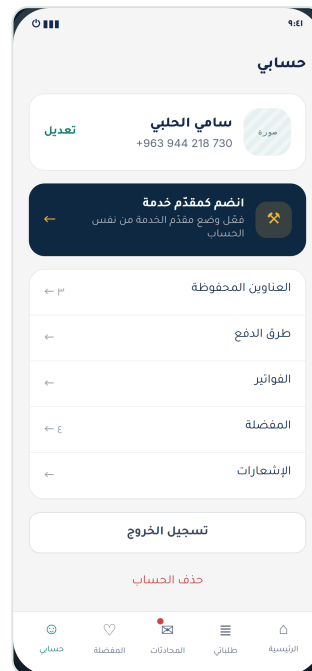
12 · Invoice
Line items and commission



13 · Rating
Multi-criteria rating



14 · My orders
History and reordering



15 · My account
Addresses and settings

The interactive prototype

The prototype is browsable screen by screen and simulates provider-side events to drive the order state — and it is what was approved before this proposal was written.

Figure 2 — The complete phase-one customer journey. Screens are taken from the approved interactive prototype at a 390 × 844 reference size.

14 Provider mode and provider dashboard

14.1 Onboarding and verification

The four verification tiers as set out in the specification, with a verification badge shown against verified accounts:

- A

Account under review
Cannot receive orders until reviewed.
- B

Registered provider
Core details have been reviewed.
- C

Verified provider
Identity and professional documents confirmed.
- D

Verified service company
Commercial registration and company details confirmed —

Phase 2

14.2 Provider dashboard sections

Section	Contents
Home	Account status · verification status · rating · new, active and completed orders · revenue · available balance · commission · response speed · acceptance rate · cancellation rate · key alerts
Orders	Available · quoted · accepted · active · completed · cancelled · stalled
Quotes	Create · edit · withdraw · quote status · expiry date
Services	Add · edit · disable · pricing model · starting price · coverage areas
Portfolio	Add photos, video and descriptions · link work to a service type · delete or edit
Wallet	Available balance · held balance · earnings · commissions · withdrawal requests · transaction log · invoices
Ratings	Average · detail · replying to a rating · reporting an abusive rating
Settings	Personal details · working hours · coverage areas · availability · notifications · language · privacy · change phone number

15 Admin panel and permissions

Module	Principal capabilities
KPI dashboard	Total customers, providers and verified providers · new, completed, cancelled and stalled orders · total order value · total commission · amounts owed · open complaints · average rating · most requested services · most active areas · user and order growth
Users	Search · details, order history, ratings and complaints · activate · suspend · ban · send a notification · edit subject to permissions · active devices and sessions
Providers	Join requests · document review · approve or reject · request further documents · verification status · specialisations · portfolio review · disable a specialisation · performance monitoring · subscriptions · commission rates · balances
Orders	View all · search and filter · details, quotes and parties · change state where necessary · manual assignment · cancel · reopen · change log · overdue order tracking
Services and categories	Main and sub categories · add a service · icon and image · pricing model · required fields · required documents · commission · minimum price · emergency availability · enable and disable · display order
Areas	Governorates · cities · districts · neighbourhoods · coverage areas · travel fees · services available in each area
Payments	Successful and failed · held · refunds · amounts owed · withdrawal requests · commissions · settlements · invoices · financial reports
Complaints	Intake · priority · assignment · corresponding with the parties · attachments · suspending a payment · recording the decision · closure · complaints report
Content	Slider · banners · information pages · FAQ · terms · privacy · tips · announcements · app messages · contact details · social links
Notifications	Send to one user or a group · by area, account type or service · scheduling · message templates
Staff	Add a staff member · create a role · set permissions · restrict financial reports · log every action · suspend or remove permissions

15.1 Pre-configured administrative roles

System administrator

General manager

Provider approvals

Customer service

Order monitoring

Complaints

Accounting

Content management

Quality

Reporting

These roles are ready-made templates, and each staff member's permissions remain individually definable from within the panel. Every administrative action is recorded in an immutable audit log capturing the actor, the timestamp, and the value before and after the change.

16 Reporting and export engine

Twenty-four reports, all **viewable, filterable and exportable to Excel and PDF**. Heavy reports are built in asynchronous jobs and the user is notified when they are ready, so the panel is never blocked.

- New users
- Providers
- Join requests
- Orders by service
- Orders by governorate
- Completed orders
- Cancelled orders
- Stalled orders
- Average order value
- Platform commission
- Provider earnings
- Payments
- Refunds
- Complaints
- Provider ratings
- Top providers
- Lowest performers
- Response times
- Quotes submitted and accepted
- Most requested services
- Most active customers
- Discount codes
- Subscriptions
- Registration sources and campaigns

The discount-code and subscription reports depend on phase-two functionality; they are delivered in phase one as ready structures that populate as soon as those features are enabled.

17 Marketing website

Content

- Introduction to the app
- Service types
- How to order a service
- How to register as a provider
- App download links
- Contact details
- Terms and conditions
- Privacy policy
- Frequently asked questions
- Contact form

Technical treatment

- Server-side rendering for load speed and search indexing
- Correct Arabic meta tags and structured data
- Page content managed from the admin panel
- Contact form protected against automated submission
- Fully responsive with RTL support
- Analytics integration

18 External integrations

An important financial distinction

The cost of building each integration — RoboAgentix's work — is entirely separate from **the provider's fees**, which the project owner pays directly to that provider. Each integration is priced as its own line in section 5 of the commercial document, as the specification explicitly requires.

Integration	Purpose	Phase	Fees
OTP service	Phone number verification	Phase 1	Per message — on the project owner
SMS service	Critical notifications	Phase 1	Per message — on the project owner
Maps and geolocation	Location, distance and arrival time	Phase 1	By consumption
Payment gateway	Electronic payment	Phase 1	Per transaction
Firebase notifications	Push notifications	Phase 1	Free within standard limits
Email service	Invoices and alerts	Phase 1	Monthly subscription
Cloud media storage	Photos, video and documents	Phase 1	By capacity and transfer
Usage analytics	User behaviour	Phase 1	Free within standard limits
Crash monitoring	Error tracking	Phase 1	Free tier sufficient at launch
Google Play and App Store	App publishing	Phase 1	Developer account fees
WhatsApp Business	An additional notification channel	Optional	Priced on request

19 Security and data protection

Requirement in the specification	Implementation
SSL-encrypted communication	TLS 1.3 enforced on every domain with HSTS, and certificate pinning in the app for sensitive paths.
Encryption of passwords and sensitive data	Argon2id hashing for any password, column-level encryption for sensitive fields, and disk encryption for backups.
Secure access tokens	Short-lived JWTs with rotating refresh tokens and reuse detection; tokens held in the device's encrypted store.
Bounded session length	A duration configurable from the panel, remote session termination, and active devices visible to the user.
Preventing repeated login attempts	Escalating rate limits by number, network address and device, with temporary lockout and an administrative alert.
Logging sensitive operations	An immutable audit log for every administrative and financial action, with actor, timestamp and before/after values.
Documents hidden from unauthorised staff	Documents in dedicated storage behind short-lived signed URLs, governed by permissions, with every access recorded.
Regular backups	Daily full backups with continuous log archiving, and documented periodic restore testing.
API protection	Mandatory authentication, strict input validation, per-route rate limiting, and prevention of over-fetching.
Protection against forged requests	Device integrity checks, signing keys for sensitive requests, and automated-pattern detection.
Fine-grained staff permissions	RBAC at action and field level, with privilege-escalation prevention and separation of duties on the financial path.
Deletion or masking under the privacy policy	Deletion on request, anonymising financial entries rather than deleting them so the ledger stays intact.

In addition: automated dependency scanning on every build, and a manual review of the critical paths — authentication, payments, permissions and file upload — before launch, with the findings addressed within the delivery scope.

20 Performance and scalability

Requirement	Implementation and target
Responsive interfaces	A target of under 400 ms API response at the ninetieth percentile on common paths, with indexes designed for every query on the critical path.
Compressing photos and video before upload	On-device compression before sending, with thumbnails and multiple resolutions generated server-side and distributed via CDN.
Progressive data loading	Cursor pagination on every list, and lazy image loading.
Not stalling on a weak connection	A local read cache, a write queue replayed when the network returns, and clear status messaging.
Absorbing user and order growth	Stateless servers behind a load balancer, horizontally scaled workers, and database read replicas when needed.

21 Reliability, backup and monitoring

Data continuity

- Daily full backups with continuous log archiving
- Point-in-time recovery
- Documented periodic restore testing
- A complete backup handed over at final delivery

Service continuity

- Fault monitoring with immediate alerting
- Centralised error and log collection
- No order loss on connection failure, via queues
- A test environment entirely separate from production

22 Testing and quality assurance

Type	Scope	Tooling and standard
Unit tests	Critical business logic	No less than 70% coverage on the financial modules and the state machine.
Integration tests	APIs and the database	Against an isolated test database created and torn down per run.
End-to-end tests	Complete user journeys	The full order cycle from creation to rating.

Type	Scope	Tooling and standard
Performance tests	Expected load	Peak scenarios against dispatch, quoting and messaging.
Security tests	Critical paths	Automated dependency scanning with manual review of authentication, payments, permissions and file upload.
Acceptance testing	The full acceptance criteria	Executing the scenarios in section 7 of the commercial document with the project owner present, signed off by minute.
Device testing	Android and iOS	A real-device matrix covering what is common in the target market.

23 Infrastructure, deployment and operations

Environment	Purpose	Specification
Development	Day-to-day team work	Local containers matching production, with seed data.
Staging	Acceptance testing and review	A production-equivalent copy with anonymised data — an explicit requirement in the specification.
Production	Live operation	Load-balanced servers, a managed database with automated backups, and object storage with a CDN.

An automated deployment pipeline: push → build → tests → security scan → deploy to staging → approval → deploy to production with immediate rollback available. Hosting, domain and external service accounts are created **in the project owner's name**.

24 Readiness for future expansion

Required expansion	Architectural readiness in phase one
New governorates and areas	The location hierarchy is data, not code — added from the panel immediately.
New countries	Currency, country code and time zone are fields in the data model from the outset.
New languages	All copy lives in translation files with internationalisation active — only translation remains.
New services	The dynamic field schema, with no code change — the core of the design.
New payment gateways	A unified payment interface — a gateway is added by writing an adapter.
Delivery companies	The job model can accept a "delivery" type within the same catalogue.
A subscription system	Plan and subscription tables are in the schema, and binding commission to a plan is ready.
Service insurance	Hook points in the order cycle for adding an insurance line and posting it financially.
A loyalty programme	The ledger accommodates points and promotional credit as entry types.
A call centre	APIs allow external systems to read and act within permission limits.
A full customer website	APIs are client-agnostic — a web interface can be built directly on top.

25 Technical deliverables at handover

- ✓ Final requirements analysis document
- ✓ Process flow diagrams
- ✓ Database design
- ✓ UX design
- ✓ An approved interactive prototype of the app interfaces
- ✓ Android application
- ✓ iOS application
- ✓ Backend system
- ✓ Admin panel
- ✓ Database
- ✓ Documented APIs
- Marketing website
- ✓ Notification integration
- ✓ Maps integration
- ✓ OTP integration
- ✓ The agreed payment integration
- ✓ Functional, security and performance testing
- ✓ App store submissions
- ✓ Complete source code
- ✓ Hosting and database credentials
- ✓ Admin panel user guide
- ✓ Training for administrative staff
- ✓ A backup at handover
- ✓ Technical documentation for the system and its APIs

RoboAgentix's undertaking

Ownership of the application and systems rests entirely with the project owner; the complete source code is handed over; and store, hosting, domain and external service accounts are created in the project owner's name. RoboAgentix may not resell the same system with the project's data or identity, nor use customer or provider data for any other purpose. The legal detail is in section 9 of the commercial document.

End of the technical document · RAX-TEC-AWNAK-001-EN · Rev. A
Read together with RAX-FIN-AWNAK-001-EN — the Commercial Proposal